

Chapter 5: Comparison of Several Mean vectors (Part 2)

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Outline

Following our approach to the one-way MANOVA, we shall start from a univariate two-way fixed-effects model and then simply generalize to the multivariate case by analogy.

Two-way ANOVA

		Factor B		no. groups	
		1	2	...	b
Factor A	1	$\{X_{11r}\}_{r=1}^n$	$\{X_{12r}\}_{r=1}^n$	...	$\{X_{1br}\}_{r=1}^n$
	2	$\{X_{21r}\}_{r=1}^n$	$\{X_{22r}\}_{r=1}^n$	...	$\{X_{2br}\}_{r=1}^n$
	$\vdots$	$\vdots$	$\vdots$		$\vdots$
no. groups g		$\{X_{g1r}\}_{r=1}^n$	$\{X_{g2r}\}_{r=1}^n$	...	$\{X_{gbr}\}_{r=1}^n$

- Observation: X_{lkr} = the r^{th} ^{for observation} observation at level l of Factor A and level k of Factor B.
- Factor A: $l = 1, \dots, g$
- Factor B: $k = 1, \dots, b$
- Replications: $r = 1, \dots, n$ For now, a **balanced** design, which means the sample sizes are equal across all factor(treatment) combinations.

$$n_A = n_B = n$$

ANOVA model for two factors

The mean of the response is modeled as

$$\underbrace{\mu_{lk}} = \underbrace{\mu}_{\text{overall level}} + \underbrace{\tau_l}_{\substack{\text{fixed effect} \\ \text{of factor A} \\ \text{at level } l}} + \underbrace{\beta_k}_{\substack{\text{fixed effect} \\ \text{of factor B} \\ \text{at level } k}} + \underbrace{\gamma_{lk}}_{\substack{\text{interaction between} \\ \text{factor A and B} \\ \text{at level } l \& k}}$$

if $\gamma = \mathbb{I}k = 0$, it is additive
if $\gamma = \mathbb{I}k \neq 0$, it is not additive

where

$$\sum_{l=1}^g \tau_l = \sum_{k=1}^b \beta_k = \sum_{l=1}^g \gamma_{lk} = \sum_{k=1}^b \gamma_{lk} = 0$$

The effects are not additive; there is an interaction.

Correspondingly, $X_{lkr} = \mu + \tau_l + \beta_k + \gamma_{lk} + \epsilon_{lkr}$.

An illustration of interactive effect

addition across A across B

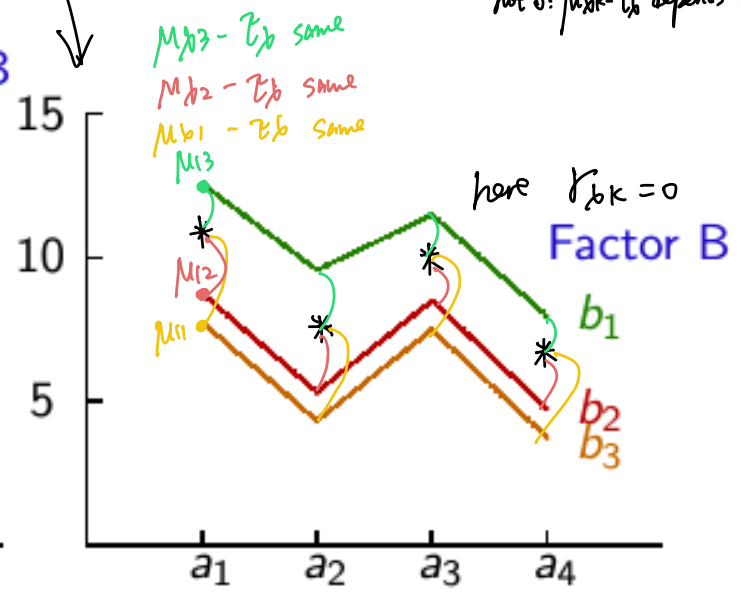
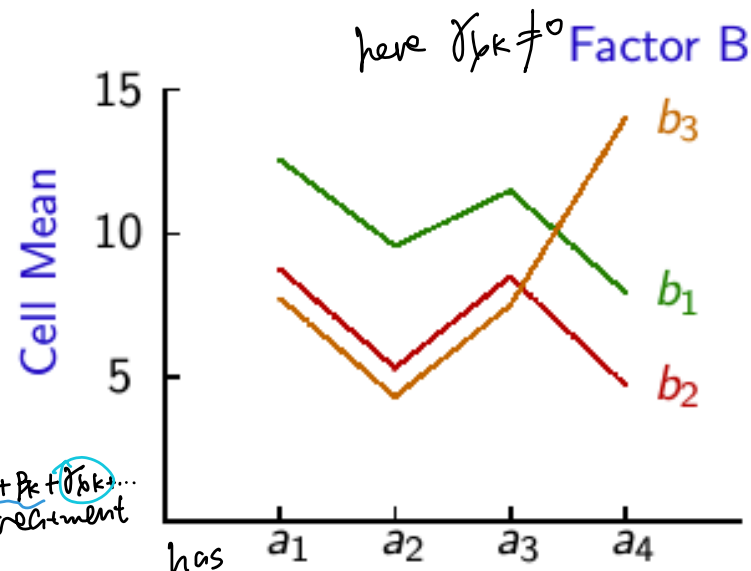
$$\mu_{AB} = \mu + \rho_A + \mu_B$$

$$\mu_{AB} - \mu_A = \mu + \rho_B$$

$$\mu_{b|k} = \mu + \tau_b + \beta_k + \gamma_{bk}$$

$$\mu_{b|k} - \tau_b = \underbrace{\mu + \beta_k}_{\text{same}} + \gamma_{bk}$$

0: $\mu_{b|k} - \tau_b$ doesn't depend on b index;
not 0: $\mu_{b|k} - \tau_b$ depends on b index.



whether the treatment has equal effect for both male and female

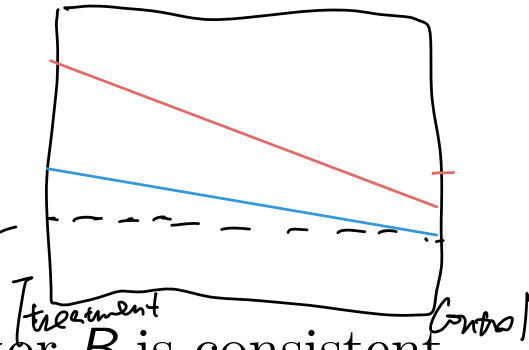
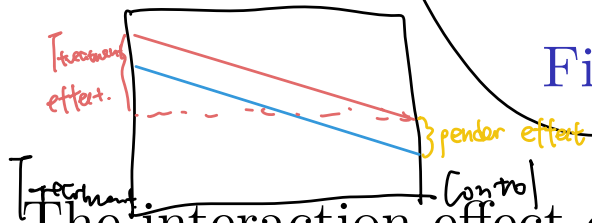
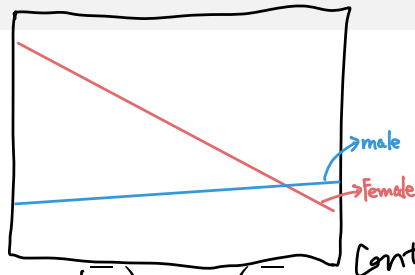


Figure: Interaction & No interaction

The interaction effect determines whether the effect of factor B is consistent across the different levels of factor A (vice versa). In other words, no interaction means that for any fixed level k of factor B , the difference $\mu_{b|k} - \tau_b = \beta_k$ should remain constant across different levels ℓ of factor A .

Observation as Sum of Means



$$X_{lkr} = \underbrace{\bar{X}}_{\hat{\mu}} + \underbrace{(\bar{X}_{l..} - \bar{X})}_{\hat{\tau}_l} + \underbrace{(\bar{X}_{.k.} - \bar{X})}_{\hat{\beta}_k} + \underbrace{(\bar{X}_{lk.} - \bar{X}_{l..} - \bar{X}_{.k.} + \bar{X})}_{\hat{\gamma}_{lk}} + \underbrace{(X_{lkr} - \bar{X}_{lk.})}_{\hat{\epsilon}_{lkr}}$$

$$\mu_{lk} = \mu + \tau_l + \beta_k + \gamma_{lk}$$

$$\mu_{lk} - \tau_l = \mu + \beta_k + \gamma_{lk} \quad (= \mu + \beta_k)$$

If $\gamma_{lk} = 0$, then $\mu_{lk} - \tau_l$ are the same for different l

For any fixed level k of factor B, the difference $\mu_{lk} - \tau_l$ should remain constant across different level l of factor A

where

- $\bar{X}$ = overall sample mean. $= \frac{1}{n \cdot g \cdot b} \sum_{l=1}^b \sum_{k=1}^g \sum_{r=1}^n X_{lkr}$
- $\bar{X}_{l..}$ = mean for l^{th} level of Factor A (or “row” mean). $= \frac{1}{nb} \sum_{k=1}^b \sum_{r=1}^n X_{lkr}$
- $\bar{X}_{.k.}$ = mean for k^{th} level of Factor B (or “column” mean). $= \frac{1}{ng} \sum_{l=1}^g \sum_{r=1}^n X_{lkr}$
- $\bar{X}_{lk.}$ = mean for the l^{th} level of Factor A and k^{th} level of Factor B (or the “cell” mean). $= \frac{1}{n} \sum_{r=1}^n X_{lkr}$
- The decomposition of the observations above shows the estimates of the population effects.

After decomposing sum of squares, we have some terms like

$$\sum_{l=1}^g \sum_{k=1}^b \sum_{i=1}^n (\bar{x}_{li} - \bar{x})(\bar{x}_{ki} - \bar{x})$$

$$= n \sum_{k=1}^b \left(\sum_{l=1}^g (\bar{x}_{lk} - \bar{x}) \right) (\bar{x}_{k.} - \bar{x}) = n \sum_{k=1}^b 0 \cdot (\bar{x}_{k.} - \bar{x}) = 0$$

Factor 1

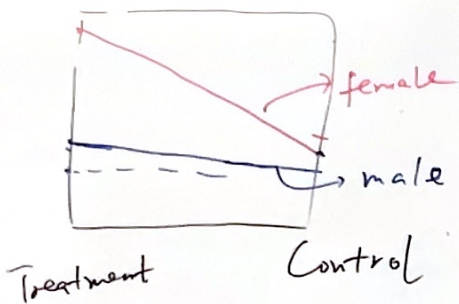
$$n \sum_{l=1}^g \sum_{k=1}^b (\bar{x}_{li} - \bar{x})(\bar{x}_{lik} - \bar{x}_{li} - (\bar{x}_{k.} - \bar{x})) = 0$$

$$= \sum_{l=1}^g \sum_{k=1}^b (\bar{x}_{lk} - \bar{x})(\bar{x}_{lik} - \bar{x}_{li}) + \sum_{l=1}^g \sum_{k=1}^b (\bar{x}_{lk} - \bar{x})(\bar{x}_{k.} - \bar{x})$$

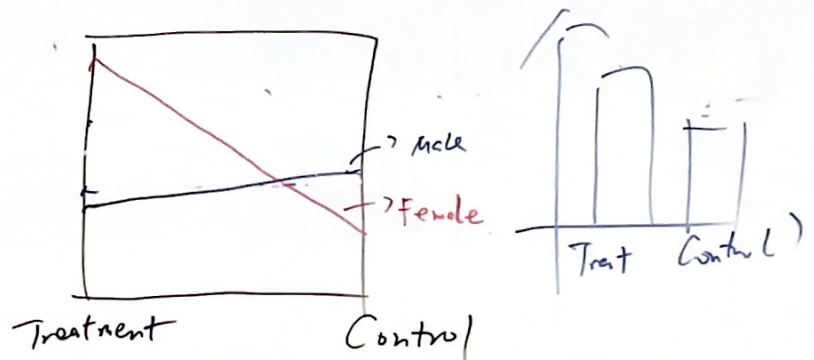
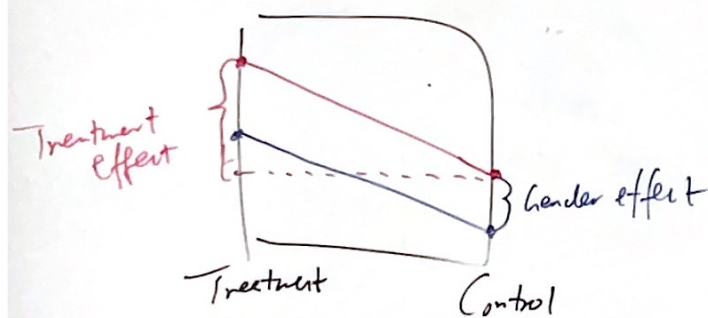
$$= \sum_{l=1}^g \sum_{k=1}^b (\bar{x}_{lk} - \bar{x})(\bar{x}_{lik} - \bar{x}_{li}) + 0$$

$$= \sum_{l=1}^g \sum_{k=1}^b (\bar{x}_{lk} - \bar{x})(\bar{x}_{lik} - \bar{x}_{li}) + 0$$

$$= 0$$



whether the treatment has equal effect for both male and female?



We cannot say treatment is effect or not in general
depend on gender

The

Decomposition of Sums of Squares

$$X_{lkr} = \underbrace{\bar{X}}_{\hat{\mu}} + \underbrace{(\bar{X}_{l..} - \bar{X})}_{\hat{\tau}_l} + \underbrace{(\bar{X}_{.k.} - \bar{X})}_{\hat{\beta}_k} + \underbrace{(\bar{X}_{l.k.} - \bar{X}_{l..} - \bar{X}_{.k.} + \bar{X})}_{\hat{\gamma}_{lk}} + \underbrace{(X_{lkr} - \bar{X}_{l.k.})}_{\hat{\epsilon}_{lkr}}$$

Handwritten notes above the equation: $\mu_{IK} = \mu_{IK} + \epsilon_{IK} + \beta_K + \gamma_{IK}$

To get the sums of squares: *all [] means Info*

[$\mu + A + B + \text{Interact}(A, B)$] - [$\mu + A$] - [$\mu + B$] + [μ]

- Subtract $\bar{X}$ from both sides; then square both sides.
- Sum over observations within cells, levels of Factor A, and levels of Factor B.

After a little algebra, The sum of squares decomposition is

$$\sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (X_{lkr} - \bar{X})^2 = \sum_{l=1}^g bn (\bar{X}_{l..} - \bar{X})^2 + \sum_{k=1}^b gn (\bar{X}_{.k.} - \bar{X})^2 + \sum_{l=1}^g \sum_{k=1}^b n (\bar{X}_{l.k.} - \bar{X}_{l..} - \bar{X}_{.k.} + \bar{X})^2 + \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (X_{lkr} - \bar{X}_{l.k.})^2$$

Handwritten red notes for degrees of freedom (df):

- $df = gbn - 1$ (for the total sum of squares)
- $df = g - 1$ (for the main effect of Factor A)
- $df = b - 1$ (for the main effect of Factor B)
- $df = (g-1)(b-1)$ (for the interaction effect)
- $df = gb(n-1)$ (for the error term)

Sums of Squares decomposition

We get an orthogonal decomposition of the sums of squares:

$$SS_{\text{total}} = SS_A + SS_B + SS_{AB} + SS_{\text{residual}}$$

Note: SS_{total} is corrected for the grand mean. Also, we get a decomposition of degrees of freedom,

$$\underbrace{(gbn - 1)}_{\text{total}} = \underbrace{(g - 1)}_{\text{Factor A}} + \underbrace{(b - 1)}_{\text{Factor B}} + \underbrace{(g - 1)(b - 1)}_{\text{Interaction}} + \underbrace{gb(n - 1)}_{\text{Residual}}$$

ANOVA summary table

Two-way ANOVA Table

Source of variation	Degrees of freedom	SS Sums of Squared
Factor A	$(g - 1)$	$SS_A = \sum_{l=1}^g bn (\bar{X}_{l..} - \bar{X})^2$
Factor B	$(b - 1)$	$SS_B = \sum_{k=1}^b gn (\bar{X}_{.k.} - \bar{X})^2$
Interaction	$(g - 1)(b - 1)$	$SS_{AB} = \sum_{l=1}^g \sum_{k=1}^b n (\bar{X}_{l.k.} - \bar{X}_{l..} - \bar{X}_{.k.} + \bar{X})^2$
Residual (error)	$gb(n - 1)$	$SS_{\text{res}} = \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (X_{lkr} - \bar{X}_{l.k.})^2$
Total	$gbn - 1$	$SS_{\text{total}} = \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (X_{lkr} - \bar{X})^2$

Hypothesis tests

- Interaction: $H_0 : \gamma_{11} = \gamma_{12} = \cdots = \gamma_{gb} = 0$. If the null is true and all assumptions valid,

$$F = \frac{SS_{AB}/((g-1)(b-1))}{SS_{\text{res}}/(gb(n-1))} \sim \mathcal{F}_{(g-1)(b-1), gb(n-1)}$$

- Factor A: $H_0 : \tau_1 = \tau_2 = \cdots = \tau_g$. If the null is true and all assumptions valid,

$$F = \frac{SS_A/(g-1)}{SS_{\text{res}}/(gb(n-1))} \sim \mathcal{F}_{(g-1), gb(n-1)}$$

- Factor B: $H_0 : \beta_1 = \beta_2 = \cdots = \beta_b$. If the null is true and all assumptions valid,

$$= \frac{SS_B/(b-1)}{SS_{\text{res}}/(gb(n-1))} \sim \mathcal{F}_{(b-1), gb(n-1)}$$

Two-way MANOVA

For p variables, $\mathbf{x}_{lkr}$ is a $p \times 1$ vector of measurements on p variables. The model

$$\mathbf{x}_{lkr} = \boldsymbol{\mu} + \boldsymbol{\tau}_l + \boldsymbol{\beta}_k + \boldsymbol{\gamma}_{lk} + \boldsymbol{\epsilon}_{lkr}$$

and $\boldsymbol{\epsilon}_{lkr} \sim \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma})$ and independent.

- $l = 1, \dots, g$
- $k = 1, \dots, b$
- $r = 1, \dots, n_{lk}$
- and for identification

$$\sum_{l=1}^g \boldsymbol{\tau}_l = \sum_{k=1}^b \boldsymbol{\beta}_k = \sum_{l=1}^g \boldsymbol{\gamma}_{lk} = \sum_{k=1}^b \boldsymbol{\gamma}_{lk} = \mathbf{0}$$

Decomposition of SS

$$\begin{aligned}
 & \underbrace{\sum_l \sum_k \sum_r (\bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}})^\top}_{SS_{\text{total}}} \\
 &= \underbrace{\sum_l bn (\bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}})^\top}_{SS_A} + \underbrace{\sum_k gn (\bar{\mathbf{x}}_{.k.} - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_{.k.} - \bar{\mathbf{x}})^\top}_{SS_B} \\
 &+ \underbrace{\sum_l \sum_k n (\bar{\mathbf{x}}_{l.k.} - \bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}}_{.k.} + \bar{\mathbf{x}}) (\bar{\mathbf{x}}_{l.k.} - \bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}}_{.k.} + \bar{\mathbf{x}})^\top}_{SS_{\text{interaction}}} \\
 &+ \underbrace{\sum_l \sum_k \sum_r (\mathbf{x}_{lkr} - \bar{\mathbf{x}}_{l.k.}) (\mathbf{x}_{lkr} - \bar{\mathbf{x}}_{l.k.})^\top}_{SS_{\text{residual}}}
 \end{aligned}$$

Tow-way MANOVA summary table

Two-way MANOVA Table

Source of variation	Degrees of freedom	SS Sums of Squared
Factor A	$(g - 1)$	$SS_A = \sum_{l=1}^g bn (\bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}})^2$
Factor B	$(b - 1)$	$SS_B = \sum_{k=1}^b gn (\bar{\mathbf{x}}_{.k.} - \bar{\mathbf{x}})^2$
Interaction	$(g - 1)(b - 1)$	$SS_{AB} = \sum_{l=1}^g \sum_{k=1}^b n (\bar{\mathbf{x}}_{l k.} - \bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}}_{.k.} + \bar{\mathbf{x}})^2$
Residual (error)	$gb(n - 1)$	$SS_{\text{res}} = \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (\mathbf{x}_{lkr} - \bar{\mathbf{x}}_{l k.})^2$
Total	$gbn - 1$	$SS_{\text{total}} = \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (\mathbf{x}_{lkr} - \bar{\mathbf{x}})^2$

Hypothesis Testing: Interaction First

most important
if $\gamma \neq 0$, then testing
main effect is meaningless

Test $H_0 : \gamma_{11} = \gamma_{12} = \dots = \gamma_{gb} = 0$ first
then $H_0 : \tau_1 = \dots = \tau_g = 0$ and $H_0 : \beta_1 = \dots = \beta_b = 0$

- Test Statistic is Wilk's Lambda

$$\Lambda^* = \frac{|SS_{\text{residual}}|}{|SS_{\text{residual}} + SS_{\text{interaction}}|}$$

Λ^* is small when $SS_{\text{interaction}}$ is large.

- Let $\nu_h = \nu_{\text{interaction}} = (g-1)(b-1)$, and $\nu_e = \nu_{\text{residual}} = gb(n-1)$
- Distribution of Λ^* are given on the following slide for various cases.
- If you don't have $p \leq 2$ or $\nu_h \leq 2$, but you have large n then the approximate distribution of

$$\left(\nu_e - \frac{p+1-\nu_h}{2} \right) \ln(\Lambda^*) \quad \text{is} \quad \chi^2_{p\nu_h}$$

Distribution of Wilk's Lambda

$$\text{Wilk's } \Lambda^* = \frac{|SS_{\text{residual}}|}{|SS_{\text{residual}} + SS_{\text{interaction}}|}$$

Number variables	<i>df</i> for Hypothesis	Sampling distribution
$p = 1$	$\nu_h \geq 1$	$\left(\frac{\nu_e}{\nu_h}\right) \left(\frac{1-\Lambda^*}{\Lambda^*}\right) \sim \mathcal{F}_{\nu_h, \nu_e}$
$p = 2$	$\nu_h \geq 1$	$\left(\frac{\nu_e-1}{\nu_h}\right) \left(\frac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim \mathcal{F}_{2\nu_h, 2(\nu_e-1)}$
$p \geq 1$	$\nu_h = 1$	$\left(\frac{\nu_e+\nu_h-p}{p}\right) \left(\frac{1-\Lambda^*}{\Lambda^*}\right) \sim \mathcal{F}_{p, (\nu_e+\nu_h-p)}$
$p \geq 2$	$\nu_h = 2$	$\left(\frac{\nu_e+\nu_h-p-1}{p}\right) \left(\frac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim \mathcal{F}_{2p, 2(\nu_e+\nu_h-p-1)}$

where ν_h = degrees of freedom for hypothesis, and ν_e = degrees of freedom for error (residual).

Hypothesis Testing: Interaction first

You should test the interaction first.

- Interpretation: The interaction effect determines whether the effect of factor A (eg. gender) is consistent across the different levels of factor B (eg. different interventions).
- If H_0 for the interaction is rejected, then you don't do tests for main effects because the Factor main effects do not have a clear interpretation.
- J&W recommend doing p-univariate 2-way ANOVAs to find out “where” the interactions exist. (i.e., are there interactions for just some of the p variables and not others?). For variables with no interaction, you can interpret main effects.
- In general, if you retain the null hypothesis for interaction, we continue and test main effects...

Hypothesis testing: Main effects

Factor A: $H_0 : \tau_1 = \tau_2 = \cdots = \tau_g = \mathbf{0}$ versus H_1 : at least one $\tau_l \neq \mathbf{0}$

$$\Lambda_A^* = \frac{|SS_{\text{residual}}|}{|SS_{\text{residual}} + SS_A|}$$

and $\nu_h = g - 1$. For $p \leq 2$ or $\nu_A \leq 2$ (i.e., $g \leq 3$), use the exact distribution given in the table.

Factor B : $H_0 : \beta_1 = \beta_2 = \cdots = \beta_b = \mathbf{0}$ versus H_1 : at least one $\beta_k \neq \mathbf{0}$

$$\Lambda_B^* = \frac{|SS_{\text{residual}}|}{|SS_{\text{residual}} + SS_B|}$$

and $\nu_h = b - 1$. For $p \leq 2$ or $\nu_B \leq 2$ (i.e., $b \leq 3$), use the exact distribution given in the table.

For large n :

$$- \left(\nu_e - \frac{p + 1 - \nu_h}{2} \right) \ln (\Lambda_h^*) \approx \chi_{p\nu_h}^2$$

Unbalanced design

The order of decomposition of SS will have an impact on the result, so this is more tricky, not it will not be considered in this course.

Example

The optimum conditions for extruding plastic film have been examined using a technique called Evolutionary Operation.

Three responses tear resistance, gloss, and opacity-were measured at two levels of the factors, rate of extrusion and amount of an additive. The measurements were repeated $n = 5$ times at each combination of the factor levels.

x_1 = tear resistance, x_2 = gloss, and x_3 = opacity							
		Factor 2: Amount of additive					
		Low (1.0%)			High (1.5%)		
Factor 1: Change in rate of extrusion	Low (-10)%	$\underline{x_1}$	$\underline{x_2}$	$\underline{x_3}$	$\underline{x_1}$	$\underline{x_2}$	$\underline{x_3}$
		[6.5	9.5	4.4]	[6.9	9.1	5.7]
		[6.2	9.9	6.4]	[7.2	10.0	2.0]
		[5.8	9.6	3.0]	[6.9	9.9	3.9]
		[6.5	9.6	4.1]	[6.1	9.5	1.9]
	High (10%)	[6.5	9.2	0.8]	[6.3	9.4	5.7]
		$\underline{x_1}$	$\underline{x_2}$	$\underline{x_3}$	$\underline{x_1}$	$\underline{x_2}$	$\underline{x_3}$
		[6.7	9.1	2.8]	[7.1	9.2	8.4]
		[6.6	9.3	4.1]	[7.0	8.8	5.2]
		[7.2	8.3	3.8]	[7.2	9.7	6.9]
		[7.1	8.4	1.6]	[7.5	10.1	2.7]
[6.8	8.5	3.4]	[7.6	9.2	1.9]		

Figure: Plastic film data

Data in R

```
> Plastic
```

	tear	gloss	opacity	rate	additive
1	6.5	9.5	4.4	Low	Low
2	6.2	9.9	6.4	Low	Low
3	5.8	9.6	3.0	Low	Low
4	6.5	9.6	4.1	Low	Low
5	6.5	9.2	0.8	Low	Low
6	6.9	9.1	5.7	Low	High
7	7.2	10.0	2.0	Low	High
8	6.9	9.9	3.9	Low	High
9	6.1	9.5	1.9	Low	High
10	6.3	9.4	5.7	Low	High
11	6.7	9.1	2.8	High	Low
12	6.6	9.3	4.1	High	Low
13	7.2	8.3	3.8	High	Low
14	7.1	8.4	1.6	High	Low
15	6.8	8.5	3.4	High	Low
16	7.1	9.2	8.4	High	High
17	7.0	8.8	5.2	High	High
18	7.2	9.7	6.9	High	High
19	7.5	10.1	2.7	High	High
20	7.6	9.2	1.9	High	High

Test interaction effects

To test for interaction, we compute

$$\Lambda^* = \frac{|\text{SS}_{\text{residual}}|}{|\text{SS}_{\text{residual}} + \text{SS}_{\text{interaction}}|} = \frac{275.7098}{354.7906} = .7771.$$

Since $p = 3$, $\nu_h = 1$, and $\nu_e = 16$, use

$$\frac{\nu_e + \nu_h - p}{p} \frac{1 - \Lambda^*}{\Lambda^*} = \frac{14}{3} \left(\frac{1 - .7771}{.7771} \right) = 1.34,$$

and it is smaller than $F_{3,14}(.05) = 3.34$. Therefore we do not reject the hypothesis $H_0 : \gamma_{11} = \gamma_{12} = \gamma_{21} = \gamma_{22} = \mathbf{0}$ (no interaction effects).

Test main effects

To test for the effect of factor A (rate of extrusion), i.e., $H_0 : \boldsymbol{\tau}_1 = \boldsymbol{\tau}_2 = \mathbf{0}$, we calculate

$$\Lambda_A^* = \frac{|\text{SS}_{\text{residual}}|}{|\text{SS}_{\text{residual}} + \text{SS}_A|} = \frac{275.7008}{722.0212} = .3819$$

Since ν_h for this test is one (factor A has two levels, $\nu_h = g - 1 = 1$), use

$$F_A = \frac{\nu_e + \nu_h - p}{p} \left(\frac{1 - \Lambda_A^*}{\Lambda_A^*} \right) \sim \mathcal{F}_{p, (\nu_e + \nu_h - p)}$$

and we find $F_A = 7.55 > F_{3,14}(.05) = 3.34$. So we reject H_0 and conclude that change in rate of extrusion affects the responses.

Conclusion

Similarly, we can test the effect of amount of additive and conclude that it also affects the responses.

The final conclusion is, both factors affect the responses, and they do so in an additive manner.